

27/11/20

Formule

$$y = f \left(\sum_{i=1}^n x_i w_i + b \right) \rightarrow \text{bias}$$

Sum of weighted inputs

$$y = (x_1 w_1 + x_2 w_2 + x_3 w_3 + \dots + x_n w_n)$$

x_i = input features

w_i = Corresponding weights

b_i = bias.

$f(\cdot)$ = activation function

y = output

② Error at output Neuron:

Actual output = y

Learning rate = η

output obtained = o

$$\delta = (y - o) \cdot o \cdot (\eta - o)$$

③ updates weights from hidden to output layer (LEI)

$$\Delta w_{ij} = \eta \cdot \delta_j + H_i$$

W_{ij} = Weight Connecting H_i neuron to H_j neuron.

η = learning rate

$\delta_i =$

(4) Error at Hidden Neurons:-

$$\delta_3 = \delta_5 \cdot w_{35} \cdot H_3 \cdot (\eta - H_3)$$

$\delta_3 \rightarrow$ error at the H_3 neuron which passes input to H_5 neuron so it's taking into consideration

(i) error at H_5 i.e. ' δ_5 '

(ii) w_{35} weight b/w H_3 and H_5 .

(iii) output of H_3 which is serving as input to H_5 .

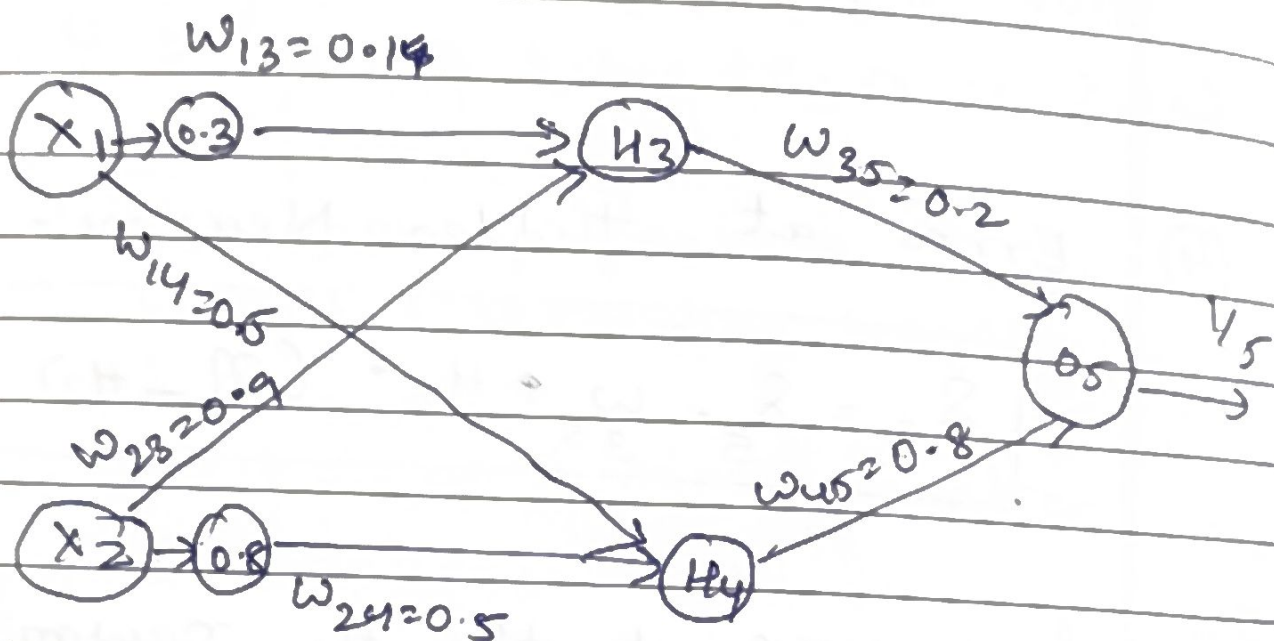
(iv) η - learning rate here is '1'

i.e. $(1 - H_3)$

where σ is the sigmoid function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Ex:-



Step ① forward bias.

Input to hidden layer

→ Inputs: $x_1 = 0.3$, $x_2 = 0.8$

Weights :

$$w_{13} = 0.1$$

$$w_{14} = 0.5$$

$$w_{23} = 0.9$$

$$w_{24} = 0.5$$

Calculate the input to the hidden neuron.

$$H_3 \neq H_4$$

$$H_3 = \sigma(x_1 \cdot w_{13} + x_2 \cdot w_{23})$$

$$H_4 = \sigma(x_1 \cdot w_{14} + x_2 \cdot w_{24})$$

where σ is the Sigmoid function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Calculations:-

$$H_3 = \sigma(0.3 \times 0.1 + 0.8 \times 0.9)$$

$$\sigma(0.03 + 0.72)$$

$$H_3 = \sigma(0.75)$$

$$H_3 = \frac{1}{1 + e^{-0.75}} \Rightarrow \frac{1}{1.47} = \boxed{0.679}$$

$$H_4 = \sigma(0.3 \times 0.5 + 0.8 \times 0.5)$$

$$H_4 = \sigma(0.15 + 0.4) \Rightarrow \frac{1}{1.57}$$

$$H_4 = \sigma(0.55) = \frac{1}{1 + e^{-0.55}} = \boxed{0.636}$$

Hidden to output layer

Weight

$$w_{35} = 0.2$$

$$w_{45} = 0.8$$

Calculate the input to the o/p neuron O_5

$$O_{5z} = (H_3 \cdot w_{35} + H_4 \cdot w_{45})$$

Calculations:

$$O_5 = (0.679 \times 0.2 + 0.634 \times 0.8)$$

$$O_5 = (0.1358 + 0.5072)$$

$$O_{5z} = (0.643)$$

$$O_5 = \frac{1}{1 + e^{-0.643}} = \boxed{0.655}$$

Step 2:- Backward Pass.

Given:

• Actual / Target output $y = 0.5$

• Learning rate $\eta = 1$

Calculate error $\Rightarrow$ update weight

Error at output Neuron.

$$\delta_5 = (y - o_5) \cdot o_5 \cdot (1 - o_5)$$

$$\delta_5 = (0.5 - 0.655) \times 0.655 \times (1 - 0.655)$$

$$\delta_5 = -0.027$$

update weights from hidden to output layer.

$$\Delta w_{36} = \eta \cdot \delta_5 \cdot H_3 \rightarrow$$

$$= 1 - 0.027 = 0.679$$

$$= -0.018$$

$$w_{35} = 0.2 - 0.018 = 0.182$$

$$\Delta w_{45} = \eta \cdot \delta_5 \cdot H_4$$

$$= 1 - 0.027 \times 0.634$$

$$= -0.017$$

$$= 0.8 - 0.017$$

$$\Delta w_{46} = 0.783$$

Error at Hidden Neurons.

$$\delta_3 = \delta_5 \cdot w_{35} + H_3 \cdot (\eta - H_3)$$

$$\delta_4 = \delta_5 \cdot w_{45} + H_4 \cdot (\eta - H_4)$$

Calculations:-

$$\delta_3 = -0.027 \cdot 0.2 \cdot 0.679 (1 - 0.679)$$

$$\delta_3 = -0.001$$

$$\delta_4 = -0.027 \cdot 0.8 \cdot 0.634 \cdot (1 - 0.634)$$

$$\delta_4 = 0.005$$

update weights from input to hidden layer

$$\Delta w_{13} = \eta \cdot \delta_3 x_1$$

$$= 1 \cdot -0.001 \cdot 0.3$$

$$= -0.0003$$

$$= 0.1 - 0.0003$$

$$\Delta w_{13} = 0.0997$$

$$\Delta w_{23} = \eta \cdot \delta_3 \cdot x_2$$

$$= 1 \cdot -0.001 \cdot 0.8$$

$$= -0.0008$$

$$= 0.9 - 0.0008$$

$$\Delta w_{23} = 0.8992$$

$$\Delta w_{14} = \eta \cdot \delta_4 \cdot x_1$$

$$= 1 \cdot -0.005 \cdot 0.3$$

$$= -0.0015$$

$$= 0.5 - 0.0015$$

$$\Delta w_{14} = 0.4985$$

$$\Delta w_{24} = \eta \cdot \delta_4 \cdot x_2$$

$$= 1 \cdot -0.005 \cdot 0.8$$

$$= -0.004$$

$$= 0.5 - 0.004$$

$$\Delta w_{24} = 0.496$$

Step 3:- Perform Another forward pass using the updated weights:-

New Weights:-

$$w_{13} = 0.0997$$

$$w_{14} = 0.4985$$

$$w_{23} = 0.8992$$

$$w_{24} = 0.496$$

$$w_{35} = 0.182$$

$$w_{45} = 0.783$$

forward Pass with updated weights.

calculate new hidden layer activation

$$H_3 = \sigma (0.3 \times 0.0997 + 0.8 \times 0.8992)$$

$$= \sigma (0.02991 + 0.71936)$$

$$= \sigma (0.74927)$$

$$H_3 \approx 0.628$$

$$H_4 = \sigma(0.3 \times 0.4985 + 0.8 \times 0.496)$$

$$= \sigma(0.14955 + 0.3968)$$

$$= \sigma(0.74927)$$

$$H_4 \approx 0.633$$

Calculate new output layer activation:

$$O_5 = \sigma(0.678 \times 0.182 + 0.633 \times 0.783)$$

$$= \sigma(0.123396 + 0.495189)$$

$$= \sigma(0.618585)$$

$$O_5 \approx \sigma 0.65$$

Thus the o/p of after performing another forward pass with the updated weights is approximately 0.65